Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1 In this question you must show detailed reasoning.**

**(a)** Sketch the curve $y = (x+2)(x-1)^2$, showing the coordinates of all the points where it meets or touches the axes. **[4]**

**(b)** Find the exact area of the finite region enclosed between the curve and the $x$-axis. **[4]**

**2** A cubic curve crosses the $x$-axis at $x = -1$, $x = 2$ and $x = 4$, and passes through the point $(0, 16)$.

**(a)** Find the equation of the curve in the form $y = k(x+1)(x-2)(x-4)$. **[3]**

**(b)** Express the equation in the form $y = ax^3 + bx^2 + cx + d$. **[2]**

**(c)** Find the set of values of $x$ for which $y > 0$. **[3]**

### Question 1

**(a)** Positive cubic. Crosses the $x$-axis at $(-2,0)$ and touches it at $(1,0)$ (a repeated root). Meets the $y$-axis where $y=(2)(1)=2$, at $(0,2)$

**(b)** $y=x^3-3x+2$. Area $=\displaystyle\int_{-2}^{1}\left(x^3-3x+2\right)\mathrm{d}x=\left[\tfrac{x^4}{4}-\tfrac{3x^2}{2}+2x\right]_{-2}^{1}=\tfrac{3}{4}-(-6)=\tfrac{27}{4}$

### Question 2

**(a)** At $(0,16)$: $16=k(1)(-2)(-4)=8k$, so $k=2$ and $y=2(x+1)(x-2)(x-4)$

**(b)** $2(x+1)(x^2-6x+8)=2(x^3-5x^2+2x+8)=2x^3-10x^2+4x+16$

**(c)** Positive cubic with roots $-1,2,4$: $y>0$ for $-1<x<2$ or $x>4$